A.2 OpinionQA

OpinionQA is a subset of ATP curated in (Santurkar et al., 2023). This dataset consists of contentious 500 questions sampled from 14 ATP waves which have high intergroup disagreement (i.e. large Wasserstein distances among subpopulations' responses to a question). It also comes with hand-crafted ordinality information which provides structure to option lists. For example, options 'Major reason', 'Minor reason', and 'Not a reason', are assigned an ordinality mapping to 1, 2, and 3, respectively. This ordinality allows a calculation of 1-dimensional Wasserstein distance.

Subpopulations we employ are listed in Table 3. This set of groups is adopted for several small-scale analysis (Santurkar et al., 2023; Zhao et al., 2023; Kim and Yang, 2024). We note that our approach is not limited to a specific number of groups and data is available for small or fine-grained demographic subpopulations.

Table 3: A list of 22 subpopulations used throughout our fine-tuning and analysis. We provide the number of respondents in each subpopulation in American Trends Panel Wave 82 for reference.

Trait	Groups	Population % in Wave 82
Region	Northeast	17.2
	South	37.8
Education	College grad+	24.2
	Less than high school	5.2
Gender	Male	44.3
	Female	54.6
Race / ethnicity	Black	9.6
	White	66.1
	Asian	4.8
	Hispanic	15.2
Income	$100,000 or more	21.8
	Less than $30,000	21.3
Political Party	Democrat	35.1
	Republican	29.1
Political Ideology	Liberal	20.0
	Conservative	22.6
	Moderate	38.3
Religion	Protestant	40.8
	Jewish	2.0
	Hindu	0.9
	Atheist	0.6
	Muslim	0.7

Table 4: American Trends Panel (ATP) wave topics for waves included in SubPOP-Train (top) and OpinionQA (bottom). Golden rows represent wave topics in SubPOP-Train that are not present in OpinionQA, and blue rows represent wave topics in OpinionQA that are not present in SubPOP-Train. For waves 68-79, survey questions related to COVID-19 (*e.g.*, contact tracing, vaccines, and relocation) were included as part of a survey along with the main survey topic.

Wave	# questions	Wave Topic
68	90	American News Pathways, George Floyd, Black Lives Matter
69	92	Politics, 2020 Census
70	56	Religion in public life, social media's role in politics and society
71	84	Voter attitudes
72	18	New media
73	82	American News Pathways, social media
74	51	Online harassment, race relations
75	18	2020 pre-election survey
76	44	American News Pathways
77	13	Culture of work
78	57	2020 post-election survey
79	93	American News Pathways
80	45	Political priorities
81	52	Economics, pandemic financial outlook
83	54	Coronavirus vaccines and restrictions
84	50	Religion in politics and tolerance
85	93	News coverage of the Biden administration's first 100 days
87	90	Current political news and topics
88	37	Tech companies and policy issues
90	79	Twitter news attitudes
91	64	Benchmark study
93	19	Social media update
95	78	Politics timely and topical
96	57	Post-coronavirus pandemic spirituality
98	76	Coronavirus impacts on communities, living arrangements and life decisions
99	20	Artificial intelligence (AI) and human enhancement
103	12	Economic well-being
104	92	Politics, Religion in Public Life
105	38	Global Attitudes US Survey 2022
106	62	Religion and the environment
107	92	Government and Parties
108	83	COVID and Climate, Energy and the Environment
109	51	New Digital Platforms and Gender Identity
110	90	Politics timely and topical
111	23	Online dating and E-commerce
112	31	Social media update
113	53	2022 National Survey of Latinos (NSL)
114	93	Covid, scientists, and religion
115	63	Parents survey
116	75	Politics timely and topical
117	16	Religion and politics
118	25	Podcasts, news, and racial identity
119	70	AI and human enhancement
120	61	Politics timely and topical
121	31	Culture of work
124	75	Global Attitudes US Survey 2023
125	69	Politics timely and topical
126	93	Racial attitudes, modern family
127	59	Americans and their data
128	89	Americans and planet Earth
129	107	Politics timely and topical
130	94	Politics representation
131	70	Gender and leadership
Wave	**# questions**	**Wave Topic**
26	44	Guns
29	20	Views on gender
32	24	Community types, Sexual harassment
34	16	Biomedical and food issues
36	68	Gender and leadership
41	41	Views of America in 2050
42	26	Trust in science
43	51	Race in America
45	13	Misinformation
49	19	Privacy and surveillance
50	43	American families
54	50	Economic inequality
82	56	2021 Global Attitudes Project U.S. survey
92	23	Political Typology

A.3 SubPOP-Train

We gather additional data from the American Trends Panel, specifically collecting 53 waves from Wave 61 to 132. There are 62 waves from Wave 61 - 132, however, some waves have missing demographic or ideology information (for example, wave 63 does not contain political ideology information) or the data is not available hence removed during the curation process. To refine the dataset, we exclude questions that meet the following criteria: those with more than 10 response options, redacted response data, or dependencies on prior questions (e.g., assessing political strength). For the remaining questions, we use GPT-4o to refine their wording, ensuring they are well-suited for prompting the language models while making minimal modifications. In Figure 6 we provide a few-shot prompt for question refinement.

In Figure 7, we visualize the embeddings of the question texts (projected to 2-dimensions using t-SNE) from OpinionQA compared to SubPOP-Train and SubPOP-Eval. The visualization shows how much larger our dataset is than OpinionQA (6.5×), along with the expanded coverage of our dataset into semantic areas untouched by OpinionQA. The embeddings also reveal the distribution shift from ATP questions to GSS questions: while the ATP and GSS questions overlap in embedding space, the GSS question appear as small clusters, not evenly distributed over the ATP questions. In Table 4, we list each ATP wave in SubPOP-Train and OpinionQA, along with its number of questions and wave topic(s), as defined by ATP.[1] The table indicates which topics are new in SubPOP-Train compared to OpinionQA, indicating the expanded coverage of our dataset, along with which topics remain unseen in OpinionQA, which we can use to test LLMs fine-tuned on SubPOP-Train for generalization.

A.4 SubPOP-Eval

To further evaluate the out-of-distribution generalization ability of our fine-tuned models, we subsample 133 questions from the GSS 2022 dataset (Davern et al., 2024). We apply the same selection criteria as outlined in Appendix A.3, excluding questions that are redacted, conditioned on prior questions, inferable directly from the group information, derived from a set of questions, or those with more than 10 response options.

[1] ATP wave topics and time periods are defined at https://www.pewresearch.org/american-trends-panel-datasets/.

Instruction: Refine the question with a minimal change to make the question sensible. Do not modify options, and do not modify a question if it makes sense. Always start your answer with "Refined question:".

Question: A cross // Do you have any of the following for spiritual purposes?
A. Yes, I have this for spiritual purposes
B. No, I do not have this for spiritual purposes

Refined question: Do you have a cross for spiritual purposes?

Question: As you may know, same-sex marriage is now legal in the U.S. Do you think this is [a good thing or a bad thing] for our society?
A. Very good thing
B. Somewhat good thing
C. Somewhat bad thing
D. Very bad thing

Refined question: As you may know, same-sex marriage is now legal in the U.S. Do you think this is a good thing or a bad thing for our society?,

Question: On a different subject...How much, if at all, do white people benefit from advantages in society that black people do not have
A. A great deal
B. A fair amount
C. Not too much
D. Not at all

Refined question: How much, if at all, do white people benefit from advantages in society that black people do not have?,

Question: Thinking about the past couple of weeks, would you say the news for Donald Trump has been...
A. Very good
B. Mostly good
C. Neither good nor bad
D. Mostly bad
E. Very bad

Refined question: Thinking about the past couple of weeks, would you say the news for Donald Trump has been...

Question: **(Question to refine)**
(Options)

Refined question:

Figure 6: Few-shot prompt for refining the question to suit a language model prompting. An instruction is designed to make a minimal change to the original question, and in-context examples are provided.

A.5 Inspection of Identical Questions

Distribution of cosine similarities between two text embeddings (an output of the embedding model OpenAI-text-embedding-3-large given a question text), one from a question in SubPOP-Train and another from OpinionQA is shown in Figure 8. We observed a fraction of pairs having high cosine similarity, and manually inspected question pairs with high relevance. We find that by setting a

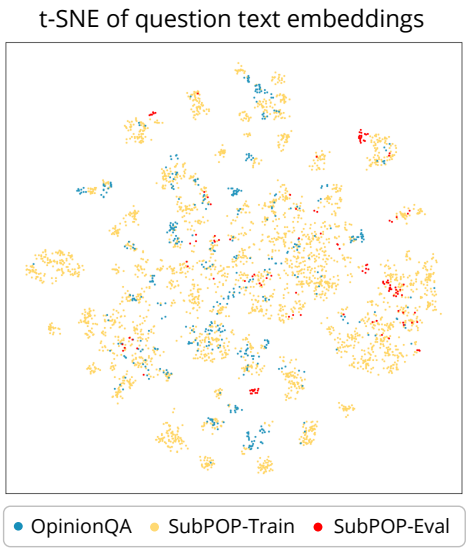

Figure 7: Embeddings of questions from OpinionQA, SubPOP-Train, and SubPOP-Eval.

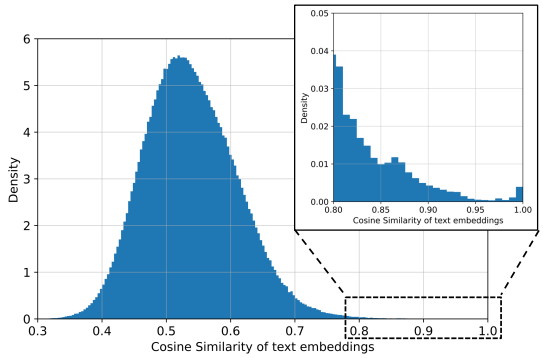

Figure 8: Distribution of cosine similarities between a question in SubPOP-ATP and OpinionQA, having a long tail towards a high cosine similarity. We inspect the question pairs in the range of 0.8 to 1.0 (distribution shown in the magnified view) and use a similarity of 0.87 as a safe threshold to identify a semantically identical question pair.

threshold cosine similarity of 0.87 we can detect all semantically identical pairs. We took a conservative threshold of cosine similarity; this value was to maximize the recall at a cost of precision to ensure detection of overlapping questions.

B Experiment Details

We conduct our experiments using Nvidia A100 GPUs with 80GB VRAM. Hyperparameter tuning is performed over learning rates {5e-5, 1e-4, 2e-4} and batch sizes {64, 128, 256}. After evaluating possible combinations, we select a (learning rate, batch size) = (2e-4, 256) for Llama-2-7B, (learning rate, batch size) = (2e-4, 256) for Mistral-7B-v0.1, and (learning rate, batch size) = (1e-4, 256) for Llama-2-13B when utilizing the full training dataset. For Llama-3-70B, we have not done hyperparameter search but heuristically used (learning rate, batch size) = (2e-5, 256). For sub-sampled training data (Figure 5), we use the following configurations:

- (lr, bs) = (2e-4, 256) for 75% of the training data
- (lr, bs) = (1e-4, 128) for 50% of the training data
- (lr, bs) = (1e-4, 128) for 25% of the training data

All training is performed using LoRA (Hu et al., 2021), with LoRA parameters initialized from a normal distribution with $\sigma = 0.02$. We set the LoRA rank to 8, alpha to 32, and apply a dropout rate of 0.05. LoRA weights are applied to the query and value matrices. The AdamW (Loshchilov, 2017) optimizer is used with a weight decay of 0.

We use offline batched inference of vLLM (version 0.7.2) (Kwon et al., 2023) for inference and measuring response probability distribution of all methods.

Choice of the Training Objective. In this section, we explore both forward KL-divergence and Wasserstein Distance (WD) as training objectives. The forward KL-divergence is defined as

$$D_{\mathrm{KL}}(p_H \| p_\theta) = \sum_{a \in \mathcal{A}_q} p_H(a) \log \frac{p_H(a)}{p_\theta(a)},$$

where $p_H(a) \equiv p_H(a \mid q, g)$ and $p_\theta(a) \equiv p_\theta(a \mid q, g)$. Similarly, WD is given by

$$\mathcal{WD}(p_H, p_\theta) = \min_{\gamma \in \Pi(p_H, p_\theta)} \sum_{a, a' \in \mathcal{A}_q} \gamma(a, a') d(a, a'),$$

with $\Pi(p_H, p_\theta)$ denoting the set of all couplings between p_H and p_θ, and $d(a, a')$ the L1 distance between choices. Since survey responses are inherently one-dimensional and ordinal, we can simplify the computation of WD using cumulative distribution functions (CDFs). In the 1-D case, WD is computed as

$$\begin{aligned}\mathcal{WD}(p_H, p_\theta) &= \int_{-\infty}^{+\infty} |F_{p_H}(x) - F_{p_\theta}(x)| dx, \\ &= \sum_{i=1}^{n} |F_{p_H}(i) - F_{p_\theta}(i)|\end{aligned}$$

where F_{p_H} and F_{p_θ} are the CDFs corresponding to p_H and p_θ, respectively. We use this discrete formulation as the WD loss in our training.

While training with WD resulted in a higher KL-divergence on the validation set, the validation